Emotional and Moral Impressions Associated with Buddhist Religious Terms in Japanese Blogs-Preliminary Analysis

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Abstract. This paper is an attempt at analyzing how much religious vocabulary (in this case Buddhist vocabulary taken from a large scale dictionary of Buddhist terms available online) is present in everyday Japanese social space (in this case in a repository of blog entries form the Ameba blog service) and thus in the consciousness of people. We also investigate and what associations (positive or negative) it generates, thus indicating the connotations associated with several Buddhist terms – whether expressions containing Buddhist vocabulary are considered proper or not from a moral point of view – as well as the emotional response of Internet users to Buddhist terminology.

Keywords: Buddhist terminology · Automated moral reasoning · Machine ethics

1 Introduction

Religion in Japan is dominated by two main doctrines – Buddhism and Shintō. According to a survey[1] carried out in 2014, less than 40% of the population of Japan identifies with an organized religion and around 45% of those are Buddhists. However, a trait characteristic for Japan is, in contrast to Western countries, a low level of identification with an organized religion paired with a high level of actual participation in religious rituals of both main religions (or even more than two religions) –the total number of people estimated to participate in some form of religious ritual according to the Japanese Ministry of Education, Culture, Sports, Science and Technology topped 190 million people, which is more than the actual population of Japan (127,171,854 people as of July 17, 2018).

It has not been yet measured to what degree is Buddhism present in the consciousness of the masses, or whether does what is considered ethical behavior according to Buddhist religious principles correspond to what the general population considers to be ethically proper or improper behavior. In the present paper we will

[1] http://www.bunka.go.jp/tokei_hakusho_shuppan/tokeichosa/shumu/pdf/h26kekka.pdf.

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analyze on a number of particular examples to what degree is religious terminology present in the vocabulary of a usual Japanese Internet user – thus what could be considered a general consciousness, and what associations does it generate, from the point of view of emotional life and moral implications.

The outline of this paper is as follows. We introduce the resources used in this study in Sect. 2, describe the method used to analyze the data in Sect. 3, the experiment for defining connotations of Buddhist terms with ethical criteria and its results are presented in Sect. 4 and the paper is concluded in Sect. 5.

2 Resources Applied in this Study

The Digital Dictionary of Buddhism[2] is the largest dictionary of Buddhist terms available online. It is a lexicon of Chinese ideograph-based terms including titles of texts, names of temples, schools, persons, etc. found in Buddhist canonical sources. As of July 20, 2018, the dictionary contains 70,814 entries.

Yet Another Corpus of Internet Sentences (YACIS, Ptaszynski et al. 2012) is a large-scale corpus of Japanese, based on blog entries from Ameba blog service.

Automatic Moral Judgement Agent Based on Wisdom of WebCrowd and Emotions (Rzepka and Araki 2005; Komuda et al. 2010) is an automatic reasoning agent applying a dynamic algorithm for moral reasoning. The model takes a sentence as an input and searches in a specified source (such as the Internet or blog corpus as the one mentioned above) for emotion types and morality-related concepts (i.e.social consequences) associating with the sentence contents. The modified phrases are queried in the corpus (e.g. the Internet) for a set specified number of snippets for one modified phrase (default is 300 per phrase). This way a large number of snippets for each queried phrase is extracted from the Web and cross-referenced with the emotive and moral lexicons. The higher hit-rate an expression had in the Web search, the stronger was the association of the original phrase to the emotion and moral types[3].

3 Initial Analysis

In the analysis we first verified how many of the headwords from the Digital Dictionary of Buddhism are present in the YACIS corpus. 22,397 terms appeared at least once in YACIS (see Fig. 1 for results). Among the terms appearing, the most popular one appeared in the corpus a total of 27,884 times, while on the other hand 3,616 least frequent terms appeared only once. A total number of 15,087,745 blog posts contained Buddhism-related terms.

It must be noted that among the terms appearing in YACIS, the majority were not strictly related to the Buddhist religion, but rather – like the term with the largest number of hits: 麻布 (jap. *azabu*, "linen", 27,884 hits) – expressions that are part of

[2] http://www.buddhism-dict.net/ddb/.

[3] For example, the phrase "thank you" is likely to associate with gratitude, relief and joy and "killing a person" is more likely to associate with such moral associations as "going to jail", or "condemn".

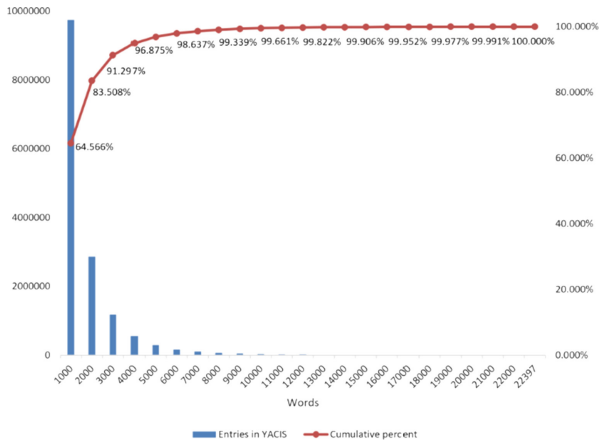


Fig. 1. 7.5% of the obtained vocabulary amounts for 83.5% of the total number of posts (the number of posts is indicated by the vertical axis)

Table 1. Top 10 terms from the Digital Dictionary of Buddhism with the greatest numbers of occurrences in YACIS corpus

Term	Reading	Meaning	Occurrences in YACIS
麻布	*azabu*	linen	27 884
時起	*Jiki*	monk name	25 391
株	*shu*	stump	24 404
胡麻	*goma*	sesame	23 713
日日	*nichi-nichi*	everyday	23 553
白	*shiro*	white	23 170
入門	*nyūmon*	to become a disciple	23 109
起動	*kidō*	move, awakening	23 015
右	*migi*	right (direction)	22 808
海水	*kaisui*	sea water	22 635

regular Japanese vocabulary, and also appear in Buddhist scriptures. Ten words with the biggest number of hits are shown in Table 1.

3.1 Phrase Filtering for Retaining Buddhist Context

After having specified which Buddhist terms appear in blogs, we needed to specify the seed phrases, which typically appear in Buddhist context, and not as lexicalized expressions. We selected the words covering 83.5% of all results, which only amounted to 7.5% of the total vocabulary found in blogs (Fig. 1). This means that the top 1,727 words generated an output of 12,070,196 posts.

We then divided the terms into those composed of one, two, three and four characters and noticed that most of the vocabulary composed of one character covered

common words, such as "egg" (卵, *tamago*, 17,842 entries), which appeared in the Buddhist canon but carried no specifically Buddhist meanings in everyday life. We decided to remove such words from the results (it accounts to a loss of only 269 words).

Proper names, such as names of Buddhist temples (in Japan usually composed of three characters, as in 吉祥寺 – *Kichijōji*, or 金閣寺 – *Kinkakuji*) or names of monks (composed of two characters, as in 一休 – *Ikkyū*, or 空海 – *Kūkai*) often appear among 2- and 3-character long words in the dictionary, but their presence in the extracted sample was negligible. Among the vocabulary composed of 4 characters, all compounds had a Buddhist meaning, one example being 自業自得 (*jigō jitoku,* "the outcome of one's own karma") – however, there were only five such words in the extracted sample.

4 Experiment

4.1 Extraction of Word N-Grams for the Investigation Emotional and Moral Associations

In the next step, we used the 12,070,196 blog posts from YACIS containing any of the 7.5% most frequent terms. The procedure was as follows: first we tokenized the sentences and lemmatized them, using MeCab[4]. Then, we extracted 3-grams and 4-grams from the sentences and retained only those n-grams that actually contained the Buddhist term. Finally we applied an additional filter to retain only those n-grams, which contained not only the word in question but also grammatical particles[5], in order to increase the proportion of samples where it appears as part of a meaningful sentence. The total number of n-grams left to consider upon performing this preliminary cleaning was 19,305,978.

4.2 Examining Ethical and Emotional Associations of Buddhist Terms

From the pool of n-grams containing Buddhist terms obtained in the previous step, we selected several representative examples and investigated their emotional and ethical associations using the Moral Judgment Agent. As a proof of concept, below we present the results for three n-gram phrases:

1. 健康を祈願する (*kenkō wo kigan suru*, "to wish/pray for health");
2. 果報を受ける (*kahō wo ukeru*, "to get what one deserves, reward or retribution");
3. 嘘をつく (*uso wo tsuku*, "to tell a lie").

[4] http://taku910.github.io/mecab/.

[5] Belonging to the categories of case markers (格助詞 *kaku-joshi*)-namely が *ga*, の *no*, を *wo*, に *ni*, へ *he*, と *to*, で *de*, から *kara* and より *yori*, adverbial particles (副助詞 *fuku-joshi*): ばかり *bakari*, まで made, だけ *dake*, ほど *hodo*, くらい *kurai*, など *nado*, なり *nari*, やら *yara*, binding particles (係助詞 *kakari-joshi*): は *wa*, も *mo*, こそ *koso*, でも *demo*, しか *shika*, さえ sae, だに *dani* and conjunctive particles (接続助詞 *setsuzoku-joshi*): ば *ba*, や *ya*, が *ga*, て *te*, のに *noni*, ので *node*, から *kara*, ところが *tokoroga*, けれども *keredomo*.

For the first term, there were only 14 results found, with social consequences annotated as "good" in all of them. However the emotional consequences (3 in total) were all "bad", with the verdict that praying for health is "an action that most people dislike" – which can be explained when presuming that prayers for health are typically done in the context of someone's illness. As a result however, the term was considered "unethical" with the confidence of 75%, while indicating that 100% of the responses found it to be "the right thing to do". At the same time, 0% of the results stipulated that it is an action "worth praise" or "in accordance with the law" – which seems contradictory.

For the second term, in the case of which connotations could be both negative and positive, a total of 253 examples were found. 100% of the 44 social and moral consequences found belonged to the positive category. There were however 25 emotional consequences, out of which only 60% were tagged as "good".

The total of both (social and emotional) factors was described using 69 emotional categories from Nakamura's dictionary; 71% of the modified phrases found indicated that this was an action worth praise and 3% further signaled that it was "the proper thing to do".

For the last term finally – one that should certainly evoke negative connotations from the standpoint of Buddhist ethics – there were 279 examples of usage found in the YACIS corpus. The total of social consequences was unexpectedly equally divided into 50% of good and bad repercussions, while the total of emotional consequences (142 instances) shifted towards "bad", with 73% versus 26%, indicating that telling a lie was considered an unethical act. In addition, 11% of the responses indicated that telling a lie was "forgivable" (vs. 3% of "unforgivable" results). There were 45 hits for telling a lie being a morally acceptable act versus 113 hits for it being unacceptable from an ethical point of view. 142 expressions of emotions were associated with telling a lie, most of which indicated it to be an act regarded as "unpleasant". The algorithm concluded that telling a lie was a definitely immoral conduct.

5 Conclusions

In this paper we presented a preliminary analysis on whether and to what degree religious vocabulary is present in the everyday of Japanese Internet users (in this case in a repository of blog entries), what is the emotional response of bloggers to Buddhist terminology and what are the results of ethical evaluations of snippets of texts including Buddhist vocabulary. In order to do this, we checked which headwords from the Digital Dictionary of Buddhism appeared in YACIS, a large-scale corpus of Japanese language. We focused on terms covering roughly 80% of all results, dividing them into 1, 2, 3 and 4 character words. We tokenized and lemmatized the blog entries containing these terms, extracted n-grams (up to 4-grams, limited to those containing the term in question and a grammatical particle) and checked a few hand-selected representative examples of expressions containing Buddhist vocabulary, using an Automatic Moral Judgement Agent Based on Wisdom of WebCrowd and Emotions (Rzepka and Araki 2005; Komuda et al. 2010)

Although a lot of headwords from the Digital Dictionary of Buddhism appeared in YACIS, a majority of those were in fact words that are part of regular Japanese

vocabulary and not specifically religious terms. Analysis of the data with the Moral Judgment Agent revealed that Buddhist terms were indeed not absent as a theme from Japanese blogs and generated a strong emotional response.

While the general reaction to several expressions using Buddhist terms was as expected, there were sometimes surprising twists in terms of social consequences. One such instance was the total of social consequences for the term uso wo tsuku ("to tell a lie"), which were unexpectedly equally divided into 50% of good and bad moral repercussions – while Buddhist ethics don't allow a concept such as a white or good lie.

As the next step in the research, we plan to develop heuristic rules in order to analyze how many words from the Digital Dictionary of Buddhism appearing in YACIS are strictly Buddhist religious terms.

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